

Def. Let X be a Metric space, Y be a Normed space.

A sequence of functions $\{f_n: X \rightarrow Y\}$ is called pointwise bounded on X if:

for any $x \in X$, $\{f_n(x)\}$ is bounded.

And it is called uniformly bounded if: for some $M > 0$,

$$\sup_X |f_n| < M \text{ for all } n \in \mathbb{N}.$$

Thm. If $\{f_n\}$ is a pointwise bounded sequence of complex-valued functions on a Countable set E , then $\{f_n\}$ has a pointwise convergent subsequence.

Proof. Denote $E = \{x_i \mid i \in \mathbb{N}\}$. Since $\{f_n(x_1)\}$ is bounded, it has a convergent subsequence, denote by $\{f_{1,k}\}$. Similarly, $\{f_{1,k}(x_2)\}$ has a convergent subsequence, denote by $\{f_{2,k}\}$. Inductively, we obtain

$$\begin{array}{ccccccc} S_1: & f_{1,1} & f_{1,2} & f_{1,3} & \dots & \downarrow \text{sub} & \text{Converges at } x_1 \\ S_2: & f_{2,1} & f_{2,2} & f_{2,3} & \dots & \downarrow \text{sub} & \text{" } x_1, x_2 \\ S_3: & f_{3,1} & f_{3,2} & f_{3,3} & \dots & \downarrow \text{sub} & \text{" } x_1, x_2, x_3 \\ & \vdots & & & & & \vdots \end{array}$$

And, the order of each functions is preserved regardless of they are contained any column, by the definition of subsequence. Then the subsequence

$$S = \{f_{n,n}\}$$

Converges for each x_i . $\square$

Note: $\mathcal{C}(X) = \{f: X \rightarrow \mathbb{C} \mid f \text{ continuous, bounded}\}$ where X is Metric space.

Lemma. If K is Compact Metric space, $f_n \in \mathcal{C}(X)$, $n \in \mathbb{N}$, and $f_n \rightarrow f$ on K , then $\{f_n\}$ is equicontinuous on K .

Proof. Using ϵ -method. $f_n \rightarrow f \rightarrow$ Conti on Compact.

Arzelà-Ascoli Theorem.

Let X be a Compact Metric space.

If $\{f_n\}$ is pointwise bounded and equicontinuous on X ,

then $\{f_n\}$ is uniformly bounded and $\{f_n\}$ contains a uniformly convergent subsequence.

Proof. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ for all $n \in \mathbb{N}$.

Since X is compact, there are finitely many points $p_1, \dots, p_r \in X$ s.t. $X = \bigcup_{i=1}^r B_\delta(p_i)$.

Since $\{f_n\}$ is pointwise bounded, there exist $M_i > 0$ s.t. for all $n \in \mathbb{N}$, $|f_n(p_i)| < M_i$.

Now, for any $x \in X$, and $n \in \mathbb{N}$, $|f_n(x)| \leq |f_n(x) - f_n(p_i)| + |f_n(p_i)| \leq \varepsilon + \max\{M_i \mid i=1, \dots, r\}$.
for some p_i s.t. $x \in B_\delta(p_i)$

To show that the second assertion, Compact + Metric $\Rightarrow$ 2nd-Countable $\Rightarrow$ separable.

Let $E \subseteq X$ be a Countable dense set. By the above Thm,

$\{f_n\}$ has a pointwise convergent subsequence on E , denote this by $\{f_{n_i}\}$. Simply, $g_i = f_{n_i}$.

Given $\varepsilon > 0$, take $\delta > 0$ s.t. $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ for all $n \in \mathbb{N}$.

Since X is compact and E is dense in X , the set

$$\{B_\delta(x) \mid x \in E\} \text{ covers } X, \text{ and}$$

has a finite subcover, $\{B_\delta(x_i) \mid i=1, \dots, m\}$. Since $\{g_i(x)\}$ converges for all $x \in X$,

there exists $N \in \mathbb{N}$ s.t. $i, j \geq N, (1 \leq i \leq m) \Rightarrow |g_i(x_i) - g_j(x_i)| < \varepsilon$.

And, for any $x \in X$, for some $1 \leq i \leq m$, $x \in B_\delta(x_i)$, thus $|g_i(x) - g_i(x_i)| < \varepsilon$ for all $i \in \mathbb{N}$.

Now, for all $x \in X$ and $i, j \geq N$,

$$|g_i(x) - g_j(x)| \leq |g_i(x) - g_i(x_i)| + |g_i(x_i) - g_j(x_i)| + |g_j(x_i) - g_j(x)| < 3\varepsilon.$$

